Conghao Li

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EDUCATION

Beijing University of Posts and Telecommunications Sep. 2023 – Jun. 2027 (expected)
B.E. in Internet of Things Engineering, International School GPA: 3.7/4.0
Dual-degree programme with Queen Mary University of London (QMUL).

Selected coursework: Advanced Mathematics I (96); Discrete Mathematics (92); Data Structures (93); Introduction to Electronic Systems (94); Digital Circuit Design (92); Machine Learning (90).

English proficiency: IELTS 6.5 (Listening 8.0; Reading 6.5; Writing 6.0; Speaking 5.5).

RESEARCH INTERESTS

LEO satellite networks; onboard computing; edge computing; semantic communications.

PUBLICATIONS & MANUSCRIPTS

*Corresponding author

1. **C. Li**, H. Yu, W. Gao*, D. Wang, S. Lai, X. Zhao, H. Zhang, and G. Liu, "QoS-Aware Downlink Paging Control for UAV-Assisted 5G-Advanced Networks with On-Demand Coverage," *Drones*, vol. 10, no. 3, Art. no. 191, 2026. doi: 10.3390/drones10030191.

2. **C. Li**, W. Gao*, W. Guo, K. Zhang, X. Chen, and H. Yu, "A Multi-Feature Fusion Approach for Dynamic User Clustering in LEO Satellite Group Handover," in *Proc. 5th Int. Conf. Frontiers of Electronics, Information and Computation Technologies (ICFEICT 2025)*, 2026. doi: 10.1007/978-981-95-8660-8_35.

3. H. Yu, **C. Li**, W. Gao*, K. Zhang, H. Zhang, and J. Shao, "Optimized Design of Multi-Layer LEO Satellite Constellation for Regional Traffic Demand in Non-Terrestrial Networks," *IEEE Trans. Veh. Technol.*, 2026. doi: 10.1109/TVT.2026.3670306.

4. S. Wan, M. Peng, H. Wang, Y. Zhang, **C. Li**, and W. Xu*, "HLFN: A Hyperspectral-LiDAR Fusion Network with LiDAR-Guided Channel-Wise Adaptation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2026, accepted.

5. H. Yu, K. Zhang, **C. Li**, W. Gao*, S. Maimaiti, and Q. Guo, "Graph-Informed Dynamic Inter-Layer Link Allocation for Multilayer LEO Satellite Networks," unpublished manuscript, 2026.

RESEARCH EXPERIENCE

Onboard Adaptive Compression of High-Resolution Satellite Images

Laboratory of Sensing and Communication for Smart Cities Shenzhen, China
Visiting Undergraduate Student, CUHK-Shenzhen Jun. 2026 – Present
Advisor: Assoc. Prof. Man-On Pun

- Develop a task-agnostic compression agent for individual satellite images, combining lightweight visual features and spatial context to select region-wise JPEG quality levels.
- Built an xView/fMoW pipeline with variable-size image tiling and actual bitstream measurement; implemented and tested a reinforcement-learning policy balancing transmission size and pixel distortion.
- *Publication plan:* Submit to IEEE ICC, followed by an extended study targeting *IEEE Transactions on Mobile Computing (TMC)*.

RESEARCH EXPERIENCE (CONTINUED)

Generative-AI Service Continuity in NTN Edge–Cloud Systems

WILL Lab, The University of Hong Kong — Hong Kong, China
Undergraduate Researcher — Jun. 2026 – Present
Advisor: Asst. Prof. Xianhao Chen

- Investigate mobility-aware state management for generative-AI services in dynamic NTN edge–cloud systems, covering conversational context, KV caches, retrieval results, tool states, and session memory.
- Model service-anchor placement and handover decisions to evaluate trade-offs among session continuity, state-migration overhead, and response latency.

Dynamic Resource Optimization for Non-Terrestrial Networks

Key Laboratory of Universal Wireless Communications (MoE) — Beijing, China
Undergraduate Researcher, BUPT — Oct. 2024 – Present
Advisors: Prof. Gang Chuai, Assoc. Prof. Weidong Gao, and Assoc. Researcher Kaisa Zhang

- Built STK–MATLAB/Python co-simulations of orbital visibility, dynamic topology, heterogeneous regional traffic, and access and handover processes in LEO and non-terrestrial networks.
- Formulated traffic-aware constellation design and inter-layer link allocation using spatio-temporal graphs, surrogate-assisted multi-objective optimization, and graph-informed marginal-gain selection.
- Designed user-clustering and QoS-aware paging methods using sparse spectral clustering, LSTM-based traffic prediction, and feedback-adaptive thresholds; benchmarked coverage, delay, blocking, energy, load balance, resilience, and signaling overhead.

Hyperspectral–LiDAR Fusion for Land-Cover Classification

State Key Laboratory of Networking and Switching Technology — Beijing, China
Undergraduate Researcher, BUPT — Sep. 2025 – Apr. 2026
Advisors: Assoc. Prof. Wenjia Xu and Prof. Mugen Peng

- Contributed to a hyperspectral–LiDAR fusion network combining spectral–spatial feature extraction, structure-aware LiDAR tokens, LiDAR-guided channel-wise adaptation, and Swin Transformer fusion.
- Conducted benchmarking and ablation studies on Houston, Trento, and MUUFL, comparing classification performance with CNN- and Transformer-based baselines.

SKILLS

- **Programming and scientific computing:** Python, C, Java, MATLAB, and HTML; C-based microcontroller programming with Keil MDK; GPIO, ADC, and UART configuration.
- **Tools and platforms:** STK, LaTeX, Git, Keil MDK, Linux development workflows, serial communication, and basic hardware debugging.
- **Research methods:** satellite orbit and link modelling, K-means and spectral clustering, evolutionary multi-objective optimization, and system-level simulation and evaluation.

HONORS & AWARDS

Honorable Mention (H Award), MCM/ICM (COMAP) — Jan. 2025
Third Prize, National College Students Mathematics Competition, Beijing Division — Nov. 2025
University-level Scholarship, BUPT.